Are separate fields converging on the same principle of recursive amplification?

Empirical work · Review paper · Paper XI

Recursive Dynamics · The ARC Theory · bearing on Law I, the ARC Principle, a law under test

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Abstract

I believe advanced AI is an existential risk unless values and correction are built into how it reasons; only proof will move the world. This paper asks how later work in separate fields bears on the ARC Theory's conjectured links between recursive capability, the limits of external control and correction embedded in a system. It presents a selected retrospective register beside the author's dated records over a 21-month window from 8 December 2024 to 1 September 2026. The contribution is a source trail, a record of revised comparisons and a separation of questions that require different measurements. The retained historical table and a frozen graded-register snapshot identify available sources and unresolved cells. They supply no aggregate evidential verdict: no fixed search population, disconfirming denominator or chance model is available. One dependent lineage connects a paper on diversity among machine agents (Hernández-Espinosa et al. 2026) with version 1 of a formal-verification preprint (Gumbau Mezquita 2026a); the text distinguishes that historical version from its revised version 4. The human parallel is argued separately in Paper C. The oversight comparison is equally specific: Unit 01, a design preregistered by timestamp, measures a matched difference in residual-error decay at fixed capability; it measures neither correction-capacity scaling nor a ceiling. P6's same-class bound remains a conjecture subject to its registered domain and conditions. Section 6 states the requirements for a later audit. The register bears on Law I of the ARC Theory and, as context only, on Law V; it tests neither.

Version 1.16 states Unit 01's status as a design preregistered by timestamp and removes notes about how version 1.15 was prepared; Appendix A records every version and correction.

Paper XI of the ARC Theory, within Recursive Dynamics: a register bearing on Law I, the ARC Principle, and, as context only, on Law V, the ARC Embedding Law, each a law under test; each row is graded for the part of the principle it bears on, no total is claimed, and the pattern is recorded as context, not confirmation (Section 9 states what the paper decides and the laws' status). Working paper, version 1.16, 1 November 2026; first published 2 July 2026; Appendix A lists every version and correction. Recorded on OSF as *Paper XI · The ARC Theory · Law I, the ARC Principle · Convergence on Recursive Amplification: a Graded Cross-Domain Register*, DOI 10.17605/OSF.IO/DC9GW. About the author, founder of Recursive Dynamics, without an institution or a doctorate: michaeldariuseastwood.com/questions/who-is-michael-darius-eastwood; his book, *Infinite Architects* (first edition, 2 January 2026): osf.io/dqm3e. Licence: CC BY 4.0.

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1 Introduction

I believe advanced AI is an existential risk unless values and correction are built into how it reasons; only proof will move the world. If control of a capable AI system were lost and could not be won back, what the system itself carries would decide what it does. The programme's position is that people keep oversight, containment and shutdown for as long as they physically can, and never give them up by choice; correction built into a system is the contingency for control lost against our will, not a reason to let go of it. The programme states the principle behind that contingency in three parts: above a threshold, recursive self-correction produces super-linear capability gains; external alignment constraints are structurally insufficient to contain it; and the corrective must be embedded, not imposed. This paper asks whether later work in separate fields bears on those parts. In plain words, its answer is a record, not a verdict. It sets later work by others against the author's dated records, the earliest a manuscript of 8 December 2024 (Eastwood 2024), row by row, with dates, source availability and the part of the question each comparison bears on. The rows bear on different parts; no row, and not the set, establishes the three together; no total is claimed; and no aggregate evidential verdict follows. A researcher choosing an oversight design can use the comparisons to distinguish residual-error reduction, corrective capacity and formal verification guarantees.

This paper belongs to the ARC Theory (the Theory of Artificial Recursive Creation), the founding theory of Recursive Dynamics, the science of systems that improve themselves, which the author founded (the founding paper). The theory is carried by five laws, each a law under test: Law I, the ARC Principle, that usable capability is a power of recursive depth, with the exponent measured rather than assumed; Law II, the ARC Co-Scaling Law; Law III, the ARC Ceiling; and Laws IV and V, the ARC Persistence Law and the ARC Embedding Law, added on 24 September 2026 and registered as laws on 15 October 2026, their mathematics in development. The laws and twenty-two propositions are registered on OSF (the theory registration), and the statement paper states the theory whole. This paper is a register bearing on Law I and, as context only, on Law V; it tests neither, and Section 9 states what it does and does not show.

The register is a selected retrospective record. It preserves a table of earlier comparisons beside a graded-register snapshot and its disclosed dependence groups (Section 4). The record can be inspected source by source; it does not provide a search denominator or establish causal independence. Section 6 states what a later audit would need. A further comparison appears in Section II.A.8 of the Eden Protocol's philosophical statement.

Our contributions are:

1. A dated register of later work by others in separate fields, with dates, source availability and the graded register's evidence classes and disclosed relationships, with unresolved source cells identified (Section 4).
2. The retained snapshot of 2 July 2026, kept whole as the paper's dated record with its superseded classes, and the rule that credits priority outward wherever the dates demand it (Section 4.2).
3. A plain statement of one dependent chain, from the manuscript of 8 December 2024 to a peer-reviewed paper and a preprint that credits it, graded as context, not confirmation, with the human parallel argued separately in Paper C (Section 5).
4. Row-level checks and the requirements for a future audit, including a fixed population, dependence assessment, a disconfirming search and a comparison model (Section 6).
5. The distinction between P6's conditional, conjectural bound and Unit 01's current residual-decay contrast, with a separately specified hypothesis about human input (Section 7).

Section 2 sets the register against related work. Section 3 states the scope and limits of the retrospective method, and Section 4 the graded register and the retained snapshot of 2 July 2026. Section 5 states the

one dependent chain and the closed loop. Section 6 states the requirements for a later audit, Section 7 the conjectures and distinct measurement axes, and Section 8 what the register means. Section 9 states what the paper does and does not show, with the programme's status; Section 10 gives its limitations and Section 11 concludes. Appendix A gives the version history and every correction, Appendix B the priority record and Appendix C the notation and a glossary.

2 Related work

2.1 The rival instrument

The work that sits closest to this programme's question, and so the paper this one should be weighed against, is Engels et al. (2025), "Scaling Laws For Scalable Oversight", arXiv:2504.18530, first posted 25 April 2025 at 17:54:27 UTC (a time taken from arXiv's record alone, arXiv Atom; a NeurIPS 2025 Spotlight Poster, confirmed on 15 August 2026 against the NeurIPS virtual page for poster 115536, as recorded in the statement paper's reference dossier). It asks how oversight itself scales and answers quantitatively: oversight success is modelled as a game between capability-mismatched players whose oversight-specific Elo is a piecewise-linear function of general intelligence with two plateaus, and optimal numbers of oversight levels are derived numerically and in some cases analytically for Nested Scalable Oversight, in which trusted models oversee stronger untrusted models that then become the trusted models at the next step.

The measurable distinction is between the questions and their instruments. Engels et al. (2025) already states a slope condition: oversight improves when the overseer's game-specific Elo rises faster with general Elo than the overseen system's ($m_G > m_H$). It recommends iterating a design until that condition holds. A composition label is not a variable in that model. The ARC programme asks whether a frozen corrector-class relationship is associated with a different residual-decay exponent under a matched protocol, and separately whether corrective capacity keeps pace with capability. Neither question establishes a universal cap on Nested Scalable Oversight. P6 of the theory registration is a conjecture with an admitted population and scale; its correction-capacity exponent is distinct both from Law II's β_C and from Unit 01's residual-decay exponent (Section 7.2).

2.2 Antecedents and near-misses

On the question. Hutter asked directly whether intelligence can explode ("Can Intelligence Explode?", Hutter 2012, arXiv, 28 February 2012, read at its source), separating "speed from intelligence explosion" and undertaking to "consider possible bounds on intelligence", augmenting the analysis of Chalmers (2010). The question itself goes back to Good (1966), and the separation of speed from intelligence explosion is at least fourteen years old. These antecedents establish that the question is older; this selected comparison is not a complete novelty search. Growth exponents of the form the programme uses are older (Jones 1995).

On impossibility. Three 2025 arXiv papers argue for limits on alignment or its control: Yao (2025b), "The Alignment Trap: Complexity Barriers" (arXiv:2506.10304, v1 12 June 2025 02:30:30 UTC, a time taken from arXiv's record alone, independently observed by the Internet Archive on 13 June 2025; cited by Spizzirri (2025), arXiv:2512.03048); Yao (2025a), "On the Mathematical Impossibility of Safe Universal Approximators" (arXiv:2507.03031, 3 July 2025, whose abstract uses the phrase "irreducible uncontrollability"); and Ball et al. (2025), "On the Impossibility of Separating Intelligence from Judgment" (arXiv:2507.07341, 9 July 2025). All three are worst-case and qualitative: measure zero, coNP-completeness, cryptographic hardness. None reports an average-case scaling exponent or a rate. The first argues that the set of safe policies has measure zero and that verifying a policy's safety is

coNP-complete; the second, that perfect, reliable control of a universal approximator is a mathematical impossibility; the third, under cryptographic hardness assumptions, that some large language models admit no efficient prompt filter and that output filtering is intractable in a natural setting. The third concludes that "safety cannot be achieved by designing filters external to the LLM internals (architecture and weights)" and argues that "an aligned AI system's intelligence cannot be separated from its judgment"; that argument from filtering intractability is consistent with this programme's architecture dependence, though it does not test it. Two of the three papers are by one sole author; to call them a wave would overstate the literature's breadth, though not the seriousness of the six-author paper.

On model similarity and oversight. Goel et al. (2025) is a direct predecessor for the error-complementarity question. Its Chance Adjusted Probabilistic Agreement measures overlap in model mistakes beyond a chance baseline. It reports judge preference for similar models, a role for complementary knowledge in weak-to-strong generalisation, and increasing similarity of mistakes with capability. Unit 01's draft instead targets a matched residual-decay contrast over repeated revisions. That difference in estimand supplies a comparison to test; it does not make the general connection between model similarity and oversight new.

The nearest structural analogue. The quantum error-correction threshold result (Aharonov and Ben-Or 2008) concerns reliable computation under a specified local noise model, including exponentially decaying correlations. Its critical physical noise rate is not a correction-capacity elasticity. The comparison is methodological, with the axes kept separate (Section 7.3).

On feedback and stability (classical). The proposition that insufficient corrective gain can cause instability has a long control-theory lineage, including the small-gain analysis of Zames (1966). An exponent comparison across capability is a distinct operationalisation under test, not a claim to have originated the general stability question. Its measurement and validity remain to be established for the systems it is intended to describe.

The nearest quantitative neighbour. Liu and Meng (2026), "Self-Correction as Feedback Control: Error Dynamics, Stability Thresholds, and Prompt Interventions in LLMs", arXiv:2604.22273 (a single source, the abstract read via the paper's own html page, 12 August 2026), recasts self-correction as a closed-loop control problem via a two-state Markov model parameterised by an Error Introduction Rate and an Error Correction Rate, and derives a directly measurable stability threshold: iterate only when ECR/EIR exceeds $\mathrm{Acc}/(1-\mathrm{Acc})$. The differential, stated wherever it is cited: theirs is a per-step rate threshold at a fixed capability level, deciding whether another iteration helps now; this programme's criterion is a scaling relation across capability, deciding whether corrective strength keeps pace as the system improves. Theirs has no scaling exponent on the corrector and no capability-growth exponent, the two quantities the programme's co-scaling criterion compares, and no architecture or composition-class term; the two models meet at zero growth, as Paper X shows. Two adjacent empirical results from the same search, each read at its source: Li (2025) (arXiv:2601.00828, first posted on 24 December 2025) reports an accuracy-correction paradox, weaker models achieving materially higher intrinsic correction rates than stronger ones, per-step evidence directionally adjacent to the same-class cap and never to be quoted as an exponent measurement; and Tsui (2025) (arXiv:2507.02778, Self-Correction Bench, first posted on 3 July 2025) reports a systematic self-correction blind spot across open models.

2.3 The field around the register

The questions this register tracks from outside now have a named home: Recursive Dynamics, the field the author founded, set out in its founding paper (version 3.5 of 24 October 2026), which argues from within what this register documents from without: its Section 3.1 tries to house the field's questions in twelve candidate host fields in their own variables, prints the dissolution map by host and argues for a coupling of measurements across those fields; that comparison is not a proof that no host field can express

it. The register and the field's founding paper are one discipline pointed in two directions; the register grades what arrived from outside, and the founding paper states what would kill the name.

On recursive creation. Smolin (1992) supplies the cited antecedent for selection-shaped universes. Version 1.14 also quoted the author's December-era notes; that quotation stays in Appendix A.6 until its dated source is cited.

3 Method

This is a selected retrospective register of works encountered across the domains named in Table 1. The edition has no reproducible search log, fixed search population, complete inclusion and exclusion record, or disconfirming denominator. Its rows therefore support comparisons between named documents; they do not estimate how often the pattern being considered occurs in the literature.

For a comparison to be checkable, the record needs the author's dated passage, the outside passage, the exact source version, the relevant dates and their timestamp classes, and the relation claimed between the passages. A related theme, a shared mathematical form and a test of the same proposition are different relations. Section 4 retains the earlier table and identifies source cells that remain unresolved. The machine-readable snapshot that accompanies this edition records the graded register exactly as it stood on 1 November 2026; it does not certify the completeness or accuracy of every row.

The manuscript of 8 December 2024 and its accompanying records are the author's dated source bundle (Eastwood 2024). A document date, a message header, a repository timestamp and a Bitcoin anchor carry different information (Appendix B). The absence of a citation to the author is not evidence of causal independence. Shared citations, organisations or intellectual lineages can reveal dependence; their absence from the available record cannot exclude it. The four disclosed groups are recorded relationships, not the result of a completed independence audit. No edition-wide verification of all message hashes, affiliations, funding relationships or earliest public disclosures is claimed here.

4 The graded convergence record

4.1 The graded register

The author maintains the graded register at https://www.michaeldariuseastwood.com/research/convergences/ (33 rows, generated 6 September 2026 and last amended 7 September 2026, under the claims-governance policy of 2 August 2026). **No numerical convergence total is published there or here.** The former summed register was assembled retrospectively, without a pre-outcome scoring rule, a fixed horizon, a base-rate model or a complete failure register, and its rows cannot legitimately be summed as one evidential quantity. The graded register instead assigns each row one primary evidence class: at the amendment of 7 September 2026, 1 formal, 2 qualified-technical, 2 engineering, 4 institutional, 6 prior-work, 12 context and 6 excluded, with the strongest rows graded moderate and the engineering rows weak. Four dependence groups are disclosed there. Two of them are the exceptions this paper names: the Hernández-Espinosa to Gumbau formal-limits chain (Hernández-Espinosa et al. 2026; Gumbau Mezquita 2026a) is one dependent lineage, not two confirmations, and the interfaith-governance rows belong to one institutional movement that predates the December 2024 anchor. The other two are a compute-governance group, covering chips, fabrication, export control and hardware enforcement, which the register records as belonging to one policy and engineering literature that substantially predates the December 2024 anchor, and a group of rows that quote the staff of one organisation, which the register records as one organisational source whose rows are not independent of each other. Independent work belongs to its

authors: a later event can show that a question matters without establishing this programme's priority, a successful prediction or the truth of its answer.

The exact graded-register snapshot read for this edition was retrieved on 1 November 2026 from https://www.michaeldariuseastwood.com/research/data/evidence-register.json. It accompanies the source package as `evidence-register-snapshot.json`, SHA-256 10235ea32150ae2c1da690b86e058fe4dd1dc23612af4489fcc1ca4687fa7ecc. Its generated and amended dates above are fields in that record, not the retrieval date. The snapshot freezes this comparison's input; its classifications are the programme's own and do not establish a population-level pattern.

4.2 The retained snapshot of 2 July 2026

Superseded snapshot, retained as this paper's dated record (2 July 2026). The table below is the register as this paper first published it, before the 2 August 2026 grading policy. Its per-row classes (PREDICTION, CONVERGENT, CONCURRENT) and its gap arithmetic are superseded by the graded register above. Rows it graded CONVERGENT now grade variously prior-work, context, institutional, engineering or excluded; its row 2 in particular (Willow) is graded prior-work, because the below-threshold result was public as a preprint on 24 August 2024, before the 8 December 2024 manuscript, with the Nature version following on 9 December 2024 (Google Quantum AI and Collaborators 2024). The author's non-awareness of that preprint when writing the December 2024 manuscript is declared on the canonical row as biographical attestation; the grade does not move on it, because the public order is unchanged by awareness, and it supplies no evidential credit to the other rows. Margins in the Gap column stated against arXiv submission dates (rows 1, 8, 9, 18, 19) carry an anchor taken from arXiv's record alone (arXiv Atom), and first public disclosure has not been separately verified for those rows.

Table 1. The register as this paper first published it on 2 July 2026, retained as its dated record. The date cell for Paper III in row 5 is corrected to February 2026; its earlier wording is retained in Appendix A.9. Its classes (PREDICTION, CONVERGENT, CONCURRENT) and its gap arithmetic are superseded by the graded register (Section 4.1); A1 is the manuscript of 8 December 2024, A2 that of 30 April 2025, and the artefact dated 2 January 2026 is the book (Appendix B).

#	Domain	Convergence	Eastwood artefact date	External date	Gap (anchor: artefact date)	Class	Primary source
1	AI Safety	Anthropic alignment faking: 78% in the RL-training condition (12% baseline)	8 Dec 2024	18 Dec 2024	**10 days**	PREDICTION	arXiv:2412.14093 (Greenblatt et al. 2024)
2	Quantum Computing	Willow: first below-threshold quantum error correction	8 Dec 2024	9 Dec 2024 (Nature; the preprint reporting below-threshold operation had been public since 24 Aug 2024)	1 day if the Nature paper is the marker; the anchor itself falls later than the preprint	CONVERGENT	Nature (2024), Google Quantum AI (Google Quantum AI and Collaborators 2024)
3	AI Reasoning	The Sequential Edge: sequential beats parallel in 95.6% of configurations	8 Dec 2024 (one-sided directional statement); comparison formalised Jan 2026	4 Nov 2025	Concurrent: they precede the formalisation; priority theirs	CONCURRENT	arXiv:2511.02309

#	Domain	Convergence	Eastwood artefact date	External date	Gap (anchor: artefact date)	Class	Primary source
4	Classical Physics	Classical time crystal via nonreciprocal forces	Dec 2024 / papers Jan 2026	22 Apr 2025 (arXiv:2504.15495 v1; published in PRL 6 Feb 2026)	135 days after A1	CONVERGENT	PRL 136, 057201 (Morrell et al. 2026)
5	Acoustics	First acoustic topological space-time crystals	Paper III (Feb 2026)	Feb 2026	Concurrent with Paper III	CONVERGENT	Newton 2, 100334 (Cell Press) [a commentary on Tong, Zhang, Li, Zhang and Qiu, Acoustic realization of monoatomic topological space-time crystals, Newton 2(2), 100304; the DOI of article 100304 was first registered on 18 November 2025] (Zhang and Yang 2026; Tong et al. 2026)
6	Acoustics	First dispersive phononic time crystals (band folding, subharmonics)	Prediction code Mar 2026 [record unidentified]	Jun 2026	~3 months after prediction code	PREDICTION	Nature Communications (Liu et al. 2026)
7	Hardware	Hardware-level ethics enforcement patents (runtime ethics gate)	30 Apr 2025 (A2)	Jan 2026 (filed Jul 2025)	3 to 9 months after A2	CONVERGENT	US 2026/0010411 + US 2026/0010780 (Odeh 2026a,b)
8	Hardware	Embedded off-switches for AI compute: deadman-switch security anchors	30 Apr 2025	9 Sep 2025	~4.5 months after A2	CONVERGENT	arXiv:2509.07637 (Petrie 2025)
9	Hardware	FlexHEG tamper-proof guarantee processors (commissioned work)	30 Apr 2025 (A2)	Jun 2025 (arXiv:2506.15093; no earlier release date by its commissioning body has been verified for this row)	on the arXiv date, a gap of ~1.5 months from A2	CONCURRENT	arXiv:2506.15093 (Petrie et al. 2025)
10	Regulation	US orders Fable 5 + Mythos 5 offline: first export-control action against a deployed frontier model	2 Jan 2026 (Chokepoint)	12 Jun 2026	~5 months after book	CONVERGENT	Harvard Law Review blog; CSA whitepaper; Reuters (Jun 2026) [exact source unresolved]
11	Geopolitics	Macron at G7 Evian: "trusted partners" compute sovereignty framing	2 Jan 2026 (Chokepoint)	19 Jun 2026	~5.5 months after book	CONVERGENT	Broadband Breakfast; CNBC G7 coverage [exact source unresolved]
12	Legislation	EU Tech Sovereignty Package (Chips Act 2.0 + CADA)	2 Jan 2026 (Chokepoint)	3 Jun 2026	~5 months after book	CONVERGENT	CNBC, 3 Jun 2026 (Virkkunen) [exact source unresolved]

#	Domain	Convergence	Eastwood artefact date	External date	Gap (anchor: artefact date)	Class	Primary source
13	Religion	Pope Leo XIV first encyclical: Babel versus Jerusalem framing for AI	30 Apr 2025 (Babylon/ Eden, A2)	25 May 2026	~13 months after A2	CONVERGENT	vatican.va (Pope Leo XIV 2026); Church Times; NY Post (May 2026) [additional source unresolved]
14	Religion / AI	Christopher Olah (Anthropic co-founder) co-presents encyclical at the Vatican	8 Dec 2024 (interfaith call, A1)	25 May 2026	~17 months after A1	CONVERGENT	NY Post, 25 May 2026; Church Times [exact source unresolved]
15	Media / AI	Jack Clark (Anthropic co-founder): "gas pedal but no brake pedal"	8 Dec 2024 (A1)	5 Jun and 19 Jun 2026	~18 months after A1	CONVERGENT	BBC World Service w3ct98k8; Euronews [additional source unresolved]
16	Venture	Recursive Superintelligence: $500M at $4B for closed-loop recursive self-improvement; convergent naming with A1's core concept	8 Dec 2024 (A1)	17 Apr 2026	~16 months after A1	CONVERGENT	Sifted; TechFundingNews (Apr 2026) [exact source unresolved]
17	Semiconductors	OpenAI + Broadcom Jalapeño inference chip (9-month tape-out, 10 GW target)	2 Jan 2026 (Chokepoint)	24 Jun 2026	~6 months after book	CONVERGENT	openai.com announcement; CNBC [additional source unresolved]
18	Mathematics / Cognition	Neurodivergent influenceability as contingent solution to alignment	8 Dec 2024 (A1)	14 Apr 2026 (preprint May 2025)	~16 months after A1	CONVERGENT	PNAS Nexus pgag076 + arXiv:2505.02581 (Hernández-Espinosa et al. 2026, 2025)
19	Mathematics	The Undecidability of AGI Alignment: Unverifiability Theorem + Trilemma	8 Dec 2024 (A1)	26 Jun 2026	**~18.5 months after A1**	CONVERGENT	arXiv: 2606.28639v1 (Gumbau Mezquita 2026a)

Row 3, stated at full strength and no further. Publication priority on the quantified finding, sequential beating parallel self-consistency in 95.6 per cent of configurations at matched compute, belongs to Sharma and Chopra (arXiv:2511.02309, 4 November 2025), credited unreservedly; the programme's compute-matched formalisation followed in Paper II (22 January 2026), designed and run before sight of their paper. The December 2024 manuscript carries a one-sided directional statement and an equation in which parallel width never had a term, so a parallel exponent near zero is a structural consequence rather than a surprise; that derivation was completed in 2026 by the operational definition of R, and the conclusion carries the date of its youngest premise. The two results are never equated: theirs measures chain refinement against parallel voting on reasoning benchmarks, Paper II estimates separate error-decay exponents for sequential reasoning and parallel sampling on mathematics problems; convergent direction, different instruments.

Register discipline: the retained table is historical, including its superseded classifications. Exact sources remain unresolved for rows 10, 11, 12, 14 and 16; additional press sources in rows 13, 15 and 17 are unidentified, and row 6 does not identify the prediction-code record. Those cells are not verified evidence in this edition. A named outlet is not a source citation. Identified works are linked to their official source. The current browsing interface is at https://www.michaeldariuseastwood.com/research/convergences/; the

evidence spine at https://www.michaeldariuseastwood.com/research/evidence-spine.html is the author's record, not a substitute for an outside work's primary source.

5 The dependent chain and the closed loop

5.1 The chain, stated plainly

The manuscript dated 8 December 2024 records the author's control-failure question and embedded-correction direction (Eastwood 2024). A later paper discusses diversity among machine agents as a response to limits on controllability (Hernández-Espinosa et al. 2026). Version 1 of a subsequent preprint credits that paper and presents formal unverifiability results (Gumbau Mezquita 2026a). The last two documents form a stated intellectual lineage. Their absence of a citation to the author does not establish causal independence, and their formal claims are not identical to the manuscript's directional claim.

5.2 Three degrees of neurodivergence

Degree 1: the researcher (Paper C, PNP). Michael Darius Eastwood. ADHD, diagnosed in adulthood; autism, formally diagnosed in October 2026. Brain that connects domains uniform thinkers don't see as adjacent. Builds a framework across domains while representing himself in complex legal proceedings. Paper C sets out the PNP (Polymathic Neurodivergent Profile) as a descriptive, non-diagnostic framework. Clinical documentation exists and is held privately. The method *is* the hypothesis.

Degree 2: the protection argument (Hernández-Espinosa et al. 2026). The paper argues from undecidability and irreducibility for limits on controllability and argues for diversity of reasoning among machine agents as a contingent response. Its publisher record gives affiliations including Oxford Immune Algorithmics, a University of Oxford spinout, King's College London, the Alan Turing Institute and the University of Tokyo. Its argument concerns machine agents. The parallel with people belongs to Paper C; this is not a clinical finding about the author. This comparison uses the publisher's abstract and the accessible preprint, not the full text of the published article.

Degree 3: the historical formal argument (Gumbau Mezquita 2026a). Gumbau Mezquita's version 1, posted on 26 June 2026, credits Hernández-Espinosa and colleagues and presents two named unverifiability theorems and a soundness-completeness-tractability trilemma. The register's historical row refers to that exact version. It is one dependent lineage and supplies context for the question, not a measurement of ARC's exponents.

Current source, checked 1 November 2026. Gumbau Mezquita's version 4, dated 14 October 2026, has the revised title *Algorithmic Unverifiability of Safety for Fixed and Recursively Self-Improving Systems* (Gumbau Mezquita 2026b). It condenses the static results into classical case analysis and distinguishes verification in a succinct finite arena from unrestricted formal domains; its finite-arena verification problem is coNP-complete and synthesis PSPACE-complete. It also studies preservation under self-modification. These are limits on universal certification under stated assumptions, not a conclusion that every system is unsafe or every bounded verification task impossible. The historical version's theorem names must not be presented as the current version's formulation. This paper makes no claim that the proofs have survived every possible refutation.

5.3 The closed loop

The expression "closed loop" describes how the author relates these records:

1. *Intellectual history.* Paper C's account of the author's reasoning can explain why he asked the question. It does not measure the hypothesised recursive mechanism.
2. *Literature context.* The register places named documents beside that question. Selection and dependence remain properties to investigate, not settled credentials of the pattern.
3. *Formal scope.* The claims of Gumbau Mezquita (2026b) stand or fall under their mathematical assumptions. A thematic connection to the author's manuscript neither proves those results nor transfers them to the ARC Theory.
4. *Traceability.* Version-pinned links and dated records let a reader inspect the comparison. A file hash identifies a file's exact contents; it does not establish the truth of a claim, the time of composition, or independent discovery.

6 Falsifiability

The present claims are comparisons between specified documents. A row fails if its source does not say what is attributed to it, if its date or version is wrong, or if the stated relationship conflates different quantities. Such a finding changes that row and any passage that relies on it. The register offers no aggregate evidential verdict, so it has no aggregate rejection threshold to report.

A later audit would need the following provisions before its results could be read as evidence about a broader pattern:

1. *Source and date checks.* Pin both passages and the source versions, distinguish their timestamp classes and resolve missing primary sources. An unlocated source stays unresolved.
2. *Dependence assessment.* Fix criteria for shared intellectual lineage and documented exposure. Classify a traceable response to the author's dissemination as engagement. Do not infer independence from a missing citation.
3. *A fixed audit population.* Declare which records are eligible, how duplicate events are grouped, and how disagreements are adjudicated before scoring. Unit 58's draft re-audit, first anchored on 12 August 2026 at block 962066, is preregistered by timestamp and awaits submission; it does not supply a chance model for this paper's selected register.
4. *Disconfirming cases and a comparison model.* Define a search population, inclusion rules, a disconfirming search and the baseline against which any excess of matches would be measured. Those elements have not been supplied here. No count, rate or chance-adjusted significance follows from the present rows.
5. *Separate theory tests.* Apply the theory registration's own operational definitions and refuters to data collected under the relevant registered design. A literature resemblance does not decide a prediction about recursive capability or corrective capacity.

The practical use of this list is to turn a promising comparison into a question that another researcher could check. Until a suitable audit is specified and run, the register remains a selected historical record.

Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/MichaelDariusEastwood/arc-scaling-challenge.

7 Framework predictions the convergence pattern points to

7.1 The eclipse

The oversight question is whether corrective capacity can keep pace with the system being corrected. Proposition P6 of the theory registration, version 1.104, conjectures a correction exponent at or below one half for same-class correctors. It is a conjecture, not a consequence of a shared substrate or of statistical independence alone. The registration requires a prospectively frozen, versioned taxonomy and an independently justified capability ratio scale. It separates scaling class, lineage, substrate, training provenance, access to external ground truth and failure-mode dependence; no one axis substitutes for another. The population remains blocked until the required taxonomy is fixed.

The route to one half additionally assumes that corrective capacity is inverse residual uncertainty, that correction channels are uncorrelated, that their number grows in proportion to capability, and that their variance and quality are comparable. These assumptions require separate measurement. Neither the register nor the material of a corrector establishes them. The registered P6 refuter requires a valid exceedance of one half replicated on a second, disjoint fault set under the same settings and the registration's precision rules. Its outcomes are REFUTED, NO COUNTEREXAMPLE FOUND AMONG N and INCONCLUSIVE; a finite survey cannot return support for this universal bound. Unit 82 tests P17's proxy for error dependence, not P6 or a ceiling value (Section 9.4).

The conditional crossover illustrates why the distinction matters. In the model, capability follows $C(R) = C_0(R/R_0)^{\alpha}$, marginal correctable burden is $B(R) = b\,dC/dR$, and correction service capacity is $A(R) = aC(R)^{\gamma}$, with positive coefficients and $0 \le \gamma < 1$. The crossover is

$$\alpha_{\text{crit}} = \frac{1}{1-\gamma}. \tag{1}$$

Here γ is the correction-capacity elasticity, denoted γ_C in the Unit 01 draft; it is not a residual-decay exponent. The value two follows conditionally when $\gamma = 1/2$. At equality the exponents tie and coefficients, delay, saturation and feedback dynamics still matter. This crossover does not by itself establish finite-horizon safety or bounded harm. The model and its empirical adequacy are separate from the literature comparisons in this paper.

7.2 What Unit 01 would measure

Unit 01, the corrector-class contrast, is a draft preregistered by timestamp: its design was first anchored on 28 August 2026 at block 964465, the draft has been revised since, and it awaits the author's submission. This paper reports no new run.

As drafted, it compares the decay of residual error over repeated revisions under same-class and cross-class correctors at fixed capability; γ_N denotes that residual-decay exponent. It estimates neither γ_C, nor the capability-growth exponent α, nor the crossover in equation (1), and it does not assume $\gamma_N = \gamma_C$. Its decision rules will be published with its registration.

Corrector class is classified, not randomly assigned: a contrast found on its roster would establish an association under its protocol, not architectural causation, a universal half-exponent bound or a ceiling verdict. The disclosed pilot points towards a null class contrast, which is not a finding about the capacity ceiling.

7.3 The objection to pre-empt

Quantum error correction makes the importance of the measurement axis concrete. The threshold result of Aharonov and Ben-Or (2008) concerns reliable quantum computation under a specified local noise model;

its stated model permits exponentially decaying noise correlations. It does not require independence in the unrestricted sense asserted in this paper's earlier wording. Error suppression as redundancy or code distance grows is not the same estimand as correction-capacity elasticity against capability. No mapping between those axes is established here, so this analogy supplies neither a proof of P6 nor a blanket exemption from a properly specified counterexample.

7.4 A programme hypothesis from the same mechanism

The programme also asks whether adding human judgement can improve correction under a matched budget and information set. Debate (Irving et al. 2018), amplification (Christiano et al. 2018), recursive reward modelling (Leike et al. 2018) and reinforcement learning from human preferences (Christiano et al. 2017) put human input in different places. Training-time preference labels are not an active human controller at each correction step. A human participant's different substrate alone establishes neither complementary errors nor a higher exponent. The testable hypothesis is that measured complementary error structure predicts better correction under a specified protocol; a human-versus-model contrast would need its own frozen taxonomy, outcomes and matched arms. No such comparison is measured by this register.

8 What this means

For the programme. The register locates questions worth separating: how capability grows, how correction scales, and what verification can guarantee under a particular formal model. Its contribution is a dated comparison of named sources and a record of revisions to those comparisons. Each row must stand on its own source. Neither a shared vocabulary nor the number of rows establishes the ARC Theory.

For the researcher. The PNP framework in Paper C describes the author's account of how he connects domains. The argument of Hernández-Espinosa et al. (2026) concerns diversity among machine agents; the human parallel remains Paper C's hypothesis. Personal history can explain why a question was asked. It cannot substitute for measuring the mechanism. The source bundle and version history let a reader inspect the dated record without treating biography or a matching file hash as scientific confirmation.

For AI safety. A researcher choosing an oversight design can use this comparison to ask a concrete question: does changing the corrector improve residual-error decay under matched resources, and does that improvement persist as target capability grows? Those are separate measurements. Engels et al. (2025) already studies oversight scaling; Goel et al. (2025) measures overlap in model mistakes; Ball et al. (2025) studies computational limits on filtering. Their different targets help specify what an ARC test would have to measure. A useful evaluation must report legitimate task completion alongside correction, rather than achieve apparent safety by preventing the task. That usefulness measure is an evaluation requirement here, not a new registered result.

9 What this paper does and does not show

9.1 What the register shows

The register sets documents beside the author's dated records, with the available source identifiers, dates and classifications. Unresolved source cells are identified in Section 4.2; the graded snapshot's classes and disclosed groups are recorded classifications, not findings of a completed audit. The author's

timestamp types are distinguished in Appendix B. The lineage from Hernández-Espinosa et al. (2026) to Gumbau Mezquita (2026a) is disclosed. The Willow row credits the outside preprint's earlier date regardless of the author's account of his own awareness (Section 4.2).

9.2 What it does not show

It does not show the structural principle, or any of its three parts. The rows bear on different parts of it: a mathematics result on unverifiability addresses whether an external constraint can be verified, and says nothing about the rate at which capability grows with depth or about where a corrective must sit; a policy development or a cultural echo addresses neither the growth rate nor the sufficiency of external constraint. Neither any row nor the set is offered as establishing the three parts together, and no total is claimed. The register is not a test statistic and yields no aggregate evidential verdict. Section 6 identifies what a later audit would require. The closed loop of Section 5.3 is intellectual history and motivation, not a measurement. The paper tests none of Laws I to V.

9.3 This paper and Laws IV and V

Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The abstract identifies the programme's account of correction embedded in a system, and Section 8 records later works that argue for limits on external control or verification, a pattern it calls context, not confirmation. In the theory registration, embedded correction is carried by P21, one of the two propositions that carry Law V, and so this paper bears on Law V as context only. It compares no correction that takes part in each revision round with correction that sees only the finished output, varies no build order, and measures no gain installed by external post-training under later capability-directed training. No result in this paper is a deciding test of P15, P12 or P21; the note of 3 October 2026 that first set this out is kept in Appendix A.4.

9.4 The programme's status

Status, as at 1 November 2026. What this programme names Laws are conjectures. Laws I to III were registered on 8 September 2026 and Laws IV and V on 15 October 2026 (version 1.103), the mathematics of Laws IV and V still in development (version 1.2, 29 October 2026). Each is a law under test: Law IV has not yet been tested, and the counted test of Law V's placement limb is Unit 83, below. The theory registration of twenty-two propositions was registered on 8 September 2026 at version 1.100 and updated on 13 September 2026 (version 1.102), 15 October 2026 (version 1.103) and 29 October 2026 (version 1.104, which adds no prediction and changes no law's statement); its terms are defined in a separate register. The first registered study, Unit 06, a classification audit that tests no law, was registered on 1 October 2026 and ran on 6 October 2026; one of its two runs could not start, so it is reported in full as administratively incomplete, with no inferential verdict. A second, Unit 83, the counted test of P21 under Law V, was registered on OSF on 22 October 2026 and ran on 27 and 28 October 2026: P21 is supported under the registered test, by the programme's own run, on the access axis, family-dependent, with its mechanism axis not evaluable and one call retried under a disclosed recovery amendment (results: DOI 10.17605/OSF.IO/98HKR). Two different axes are named here. The access axis is what Unit 83 varied: both reviewers are called each round, but the corrector sees the earlier kept entries with a marked new arrival and may revise it, while the monitor sees its accumulated register with the new arrival unmarked and can only accept or reject it. The performance axis is what it measured, the outcome, as against the mechanism axis, which was not evaluable. Law V, the ARC Embedding Law, remains a law under test: its placement limb, P21, is supported on the performance axis only, with its mechanism axis not

evaluable, which is not support for the information-asymmetry mechanism; its build-order limb, read on P12, is undecided, since P12 is untested. Unit 83 is not this paper's test. A third, Unit 82, the counted test of P17, was registered on OSF on 27 October 2026 and has not yet run. The re-audit (unit 58), the boundary (unit 02, study-k) and the correction exponent (unit 05, study-ad), with the corrector-class contrast of Section 7.2 (unit 01), are drafts preregistered by timestamp, awaiting the author's submission. The planned measurements have different scopes, stated in Section 7; no established-law standing is claimed. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

10 Limitations

The register's limitations are stated where each arises; they are gathered here.

- The former summed register was assembled retrospectively, without a pre-outcome scoring rule, a fixed horizon, a base-rate model or a complete failure register; the graded register replaces its sum with a grade for each row, and its rows still cannot be summed (Section 4.1).
- No disconfirming search with its denominator has been reported, and no population, inclusion rules or base-rate model have been declared, so the register cannot yet be shown to be more than a selected sample of the domains read (Section 6).
- The rows are graded within the programme, by its own record, not by an outside reviewer; Unit 58's re-audit is a draft preregistered by timestamp, awaiting submission; its draft does not provide a chance model for this register (Section 6).
- Margins stated against arXiv submission dates (rows 1, 8, 9, 18 and 19 of Table 1) rest on arXiv's record alone, and first public disclosure has not been separately verified for those rows (Section 4.2).
- Several press-sourced rows lack an identified article or official address; the corresponding source cells are explicitly unresolved (Section 4.2).
- The author's non-awareness of the Willow preprint when he wrote the December 2024 manuscript is biographical attestation, and the grade does not move on it (Section 4.2).
- The closed loop's first two properties are intellectual history and motivation, not measurements of the framework's dynamics (Section 5.3).
- The Unit 01 draft reports pilot evidence pointing towards a null residual-decay class contrast, which is not a measurement of the capacity ceiling (Section 7.2).

11 Conclusion

This selected retrospective register places named later works beside the author's dated records and preserves how those comparisons changed. It supplies a source trail and a set of questions for further work, not an aggregate evidential verdict. The dependent chain is version-pinned, unresolved source cells remain visible, and the human parallel belongs to Paper C. The oversight question is concrete: whether a change in correction improves performance under a fixed protocol, and whether correction capacity then keeps pace with capability. The current Unit 01 draft addresses the first through a residual-decay contrast; it cannot decide the second or a universal ceiling. Law I remains a law under test, and the theory registration's refuters govern any verdict on the theory.

Code and data. The graded-register snapshot read for this edition is included as `evidence-register-snapshot.json`, with its hash in Section 4.1. The browsing interface is published at

https://www.michaeldariuseastwood.com/research/convergences/, and the evidence spine at https://www.michaeldariuseastwood.com/research/evidence-spine.html; both are the author's own records (Section 4.2). The manuscripts of 8 December 2024 and 30 April 2025, and the message headers of 8 December 2024, are deposited on OSF (doi.org/10.17605/OSF.IO/QASRK). The paper and its earlier versions are deposited at doi.org/10.17605/OSF.IO/DC9GW, and its files are in the public repository github.com/MichaelDariusEastwood/arc-principle-validation, folder `papers/Paper-XI-Convergence`.

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G. Zames. On the input-output stability of time-varying nonlinear feedback systems Part one: Conditions derived using concepts of loop gain, conicity, and positivity. *IEEE Transactions on Automatic Control*, 11(2), 1966. doi:10.1109/TAC.1966.1098316. Pages 228 to 238.

L. Zhang and Y. Yang. Building a space-time crystal with sound. *Newton*, 2(2), 2026. doi:10.1016/j.newton.2025.100334. Article 100334; a commentary on Tong et al. (2026).

A Version history and corrections

A.1 Version 1.16

Version 1.16, 1 November 2026. Substantive. Section 7.2 now describes Unit 01 as a draft preregistered by timestamp, first anchored on 28 August 2026 at block 964465, as Section 9.4 already did, and leaves the draft's decision rules to its registration; the abstract and the glossary follow it. Where version 1.15 described how it had been checked, the text now states the evidence: the December-era quotation on Smolin stays in Appendix A.6 until its dated source is cited, and Section 5.2 names the sources its comparison uses. Sections 3 and 5.3 describe the register snapshot and file hashes in plain words. The entry for version 1.15 now records its change of title and its withdrawal of version 1.14's assurance on source verification, and neither it nor the closing statement of version 1.14 describes how version 1.15 was prepared. No result, registration, source or date of a claim is added or moved, and the first-publication date remains 2 July 2026.

A.2 Version 1.15

Version 1.15, 1 November 2026. Substantive. The title is now a question, *Are separate fields converging on the same principle of recursive amplification?*; until this version it read *Paper XI · The ARC Theory · Law I, the ARC Principle · Convergent Evidence for Recursive Amplification: a Graded Cross-Domain Register*. The paper adopts the programme's common typesetting form and states its contribution as a selected retrospective register. It supplies the exact graded-register snapshot read for this edition, identifies unresolved sources beside the retained table, and withdraws unsupported assurances of systematic search, causal independence, aggregate evidential force and a verifiable primary source for every row shown. The historical Gumbau preprint, version 1, is distinguished from version 4's current formulation. The comparison with Engels and colleagues recognises its slope condition; Goel and colleagues is added as nearest work on error similarity and oversight. P6's conjectural status, admitted domain and conditions are stated from the theory registration. Unit 01's current draft is described as a residual-decay contrast, with a gated secondary ratio and no capacity-ceiling verdict. The quantum threshold result's noise assumptions are corrected. Laws IV and V and the programme's registered results retain their current stated status. The author's personal certification and standard disclosure are preserved. Two bibliography version notes and the concluding statement of version 1.14 are kept, the latter as labelled history. Earlier wording remains below as the record of what was printed, not as current findings. Missing source details are marked unresolved rather than silently supplied. The first-publication date remains 2 July 2026.

A.3 Earlier versions, as printed

Each entry is printed as it was published; section numbers in these entries are those of the version they describe, and Table 2 gives where each part now stands.

Version 1.14 (5 October 2026). Substantive. The title now follows the programme's uniform order (paper, its law or framework, its own title) and reads *Paper XI · The ARC Theory · Law I, the ARC Principle · Convergent Evidence for Recursive Amplification: a Graded Cross-Domain Register*; until this version it read *Convergent Evidence for Recursive Amplification: a Graded Cross-Domain Register Bearing on Law I (Paper XI)*. The plate beneath the author line now gives the paper's place in the programme in the words of the programme's map. Mentions of the programme's papers in this paper now carry their uniform titles. §5 and the references now cite the founding paper at version 3.4 of 5 October 2026, the version that carries its uniform title, in which objection A12 is unchanged; the reference entry no longer

repeats the paper's designation after its title. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Version 1.13 (4 October 2026). Substantive. The title now names the law the register bears on, Law I, and the heading carries the same title; until this version the paper was titled *Convergent Evidence for Recursive Amplification*, under the heading *Convergent Evidence for Recursive Amplification as a Cross-Domain Structural Principle*. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Version 1.12 (4 October 2026). Substantive. A dated note places this paper against Laws IV and V, which were added to the theory on 24 September 2026 and are neither registered as laws nor tested, their mathematics set out at version 1.0 of 3 October 2026. The note says that the paper bears on Law V as context only, and that no result in it is a deciding test of P15, P12 or P21. The epistemic-status line now gives the standing of Laws I to III and of Laws IV and V separately. §5 names Recursive Dynamics as the field the author founded and calls the paper that sets it out the field's founding paper. In §2, Degree 2 now states the PNAS paper's claim in the terms of that paper's own abstract. Four reference entries now print their titles in full as their registries give them (Paper C, Infinite Architects, the PNAS paper, and the first version of the Gumbau paper, whose later versions carry other titles), and the entry for the manuscript of 8 December 2024 says that it is held privately and has no online source. The abstract now describes the PNAS paper and the Gumbau preprint in the terms of their own abstracts. In §2, Degree 2 is headed The Protection Argument, and Degree 3 names the preprint's two theorems and their corollary as its first version does. The draft registrations named in the abstract, §6 and §8 are given as drafted, preregistered by timestamp, with the date of the first anchor, and awaiting human submission. The founding paper is cited at version 3.1 of 3 October 2026. Working labels in §1 and §5 and the word operator in §5 are now plain words. Eighteen works the text cites now have reference entries; six of them are the commentary and the research article of row 5 of the retained snapshot and the four works that its rows 7 to 9 cite. Row 5 links the identifier it prints, that of the commentary, and a bracket names the research article the commentary reports on and gives the day that article's DOI was first registered, 18 November 2025, which is earlier than both dates the row prints; the row's printed words stand. The entry for Pope Leo XIV's encyclical gives the day it was signed, 15 May 2026, beside the day it was presented, 25 May 2026. A second dated note records what was printed before. The works the paper cites are linked to their official sources wherever one was confirmed. The version surfaces and the modification dates move to this revision. Beyond the bracket in row 5 and the links, no row, class, grading or refutation condition changes, and the first published date of 2 July 2026 does not move. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Correction note (3 October 2026). Until this version, the paragraph above described Recursive Dynamics as proposed as a field in its founding paper, and called that paper the field proposal. It now names Recursive Dynamics as the field the author founded, and the paper as the field's founding paper. The epistemic-status line beneath the citation block said that what this programme names Laws are conjectures under registered adversarial test. Laws IV and V, added to the theory on 24 September 2026, are neither registered as laws nor tested, and the line now gives their standing apart from that of Laws I to III. In §2, Degree 2 said that Gödel/Turing prove external alignment is impossible. In its own abstract, the PNAS paper offers a proof that misalignment is inevitable and that full orchestrated controllability of agentic systems is impossible, and Degree 2 now says that. In the references, the entries for Paper C, for Infinite Architects, for the PNAS paper and for the Gumbau paper printed short forms of their titles. They now print the titles in full as their registries give them, the Gumbau paper's as the title of its first

version. The entry for the manuscript of 8 December 2024 now says that it is held privately and has no online source. The abstract called the PNAS paper the independent paper proving that such cognition is a peer-reviewed argument for engineered cognitive diversity among AI agents, and called the Gumbau paper formal unverifiability results in a preprint. It now describes each in the terms of its own abstract. In §2, Degree 2 was headed The Protection Proof: the PNAS paper offers a proof that misalignment is inevitable, and argues for diversity among agents, so the heading now says The Protection Argument. Degree 3 named the Gumbau paper's two theorems as Unverifiability of Alignment and the Trilemma, and now names them as its first version does, with the Trilemma as their corollary. The abstract, §6 and §8 said of their draft registrations only that they await human submission; each is now also given as preregistered by timestamp, with the date of its first anchor. §5 and the references cited the founding paper at version 2.8; they now cite version 3.1 of 3 October 2026, in which objection A12 is the same objection. §1 and §5 printed working labels in capitals for how a source was read or dated, and §5 called the author's notes the operator's notes; both are now in plain words. In the retained snapshot, row 5 printed the identifier of a commentary without a link; the identifier is now linked, and a bracket names the research article that the commentary reports on and gives the day that article's DOI was first registered, 18 November 2025. That day is earlier than both dates the row prints, the date of the external work and the date of Paper III. Eighteen works that the text cites and the reference list lacked now have entries, each from its record; six of them are that commentary, that research article and the four works that rows 7 to 9 of the snapshot cite. The entry for Pope Leo XIV's encyclical gave only 25 May 2026, the day it was presented; it now also gives 15 May 2026, the day it was signed and the date the Holy See's own page gives it. No result, date or limit changes.

Version 1.11 (21 September 2026). Substantive. In the retained pre-grading snapshot, row 4 (the classical time crystal via nonreciprocal forces) now states the external work's first public date, 22 April 2025, the submission date of arXiv:2504.15495 v1, and its gap reads 135 days after A1. The fourteen-month figure the row had printed is withdrawn: the canonical evidence register withdrew that gap claim on row EVR-10 on 2 August 2026, and the snapshot row had not been carried with it. Row EVR-10 is corrected at this same revision, from the month-precise 1 April 2025 and 114 days to the day-precise 22 April 2025 and 135 days. The Physical Review Letters publication of 6 February 2026 stands as the publication date and the row now states it as such; the row keeps its CONVERGENT class, superseded, as every class in this snapshot is, by the graded register above. As with the other margins this snapshot states against arXiv submission dates, first public disclosure has not been separately verified for this row. No other row, class, date, grading or refutation condition changes, the retained snapshot keeps its rows and its superseded label, and the first published date of 2 July 2026 does not move.

Version 1.10 (20 September 2026). Substantive. The graded register's size and shape are corrected to the register as it now stands, thirty-three rows, generated 6 September 2026 and last amended 7 September 2026, in the abstract, in §1, in the page metadata and in the structured data; §1's class tally is corrected to 1 formal, 2 qualified-technical, 2 engineering, 4 institutional, 6 prior-work, 12 context and 6 excluded; §1 now states that the register discloses four dependence groups, names the two further ones and gives the register's basis for each, and the reading key's count of disclosed dependence groups is corrected with it; the count of source-pending candidate rows is withdrawn, because the published register carries no candidate list; the clause that reader engagement is logged separately is scoped to the rule it states, because no engagement list is published; and the pointer that promised quarantine and engagement lists is corrected to the register itself and to what the evidence spine carries. Six dated notes are attached, one beneath the abstract, one in the reading key, two in §3, one in §6 and one in §7, each in the wording of the owner review of 19 September 2026: they scope the opening claim to what the rows are graded to carry,

align the register's evidential role with the tests §6 records as outstanding, class the first two closed-loop properties as intellectual history rather than measured outcomes, apply §1's comparison standard to the third, and carry the defeat threshold's narrowing into the rate language. Two further notes, drafted for §2 and §4, carry a bracketed clause that only the author can settle, whether an inspection record was kept and whether the search record was kept, and neither publishes at this revision because neither bracket is answered. Editorial. The preregistration banner cites the operational definitions companion at its current version, 1.8.0 of 7 September 2026; the Recursive Dynamics founding paper is cited at v2.8; and the version surfaces and the modification dates move to this revision. No section, row or table is removed. No row, no class, no date, no grading and no refutation condition changes; the retained pre-grading snapshot keeps its rows and its superseded label; and the first published date of 2 July 2026 does not move.

Version 1.9. The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

Version 1.8 (2 September 2026). The abstract now names two disclosed exceptions, matching the two dependence groups set out in Section 1; in the retained snapshot the Willow row carries both of its dates (preprint 24 August 2024, Nature 9 December 2024), and the FlexHEG row is dated to the month of its arXiv posting; the claim that the three groups were unaware of one another is corrected, since the Gumbau paper builds on Hernández-Espinosa's; the PNAS paper's argument reads as one about machine agents, with the human parallel attributed to Paper C; the Gumbau theorems carry a date and are described as unrefuted rather than proved; the defeat-threshold paragraph is reconciled with the no-total policy; the ceiling reads as the reciprocal of the shortfall, $\alpha_{\text{crit}} = 1/(1-\gamma)$, replacing the retracted reciprocal-of-the-exponent form the text had printed; the Spotlight status of Engels et al. is verified; and the references are corrected (Greenblatt et al. for the alignment-faking paper, both Willow dates, Paper X's current title).

Version 1.7 (1 September 2026). Dating convention and register alignment.

A.4 Notes moved from the body, as printed

These notes stood in the body of earlier versions, each beside the passage it governed. Each passage is now set in the scoped form its note gives, and the note is kept here as printed.

Scope note (19 September 2026). An owner review of this opening finds that it states more than the graded register is built to carry, and this note scopes it to what the rows establish. The same paragraph declares the register's own taxonomy: technical experiments, theorems, engineering proposals, policy developments, cultural echoes, prior work and author-run studies cannot legitimately be summed as one evidential quantity, and no total is published. Rows that are not commensurable enough to be summed are not commensurable enough to have reached the same three-part principle between them. The rows bear on different components of it: a mathematics result on unverifiability addresses whether an external constraint can be verified, and says nothing about the rate at which capability grows with depth or about where a corrective must sit; a policy development or a cultural echo addresses neither the growth rate nor the sufficiency of external constraint. Each row is accordingly graded on its own, for the component it bears on, and neither any single row nor the set as a whole is offered as establishing the conjunction of the three. Unchanged: the window and its dates; every row, its class and its source; the no-total policy; the two exceptions this paper names, the Hernández-Espinosa to Gumbau chain and the interfaith-governance

rows; and the statement that no group cites another and no group cites the author's work, which is the documentary form.

Reading key. This paper carries a *result*: the graded convergence register, in which each independently sourced observation is classified and evidence-graded rather than summed into a single number. Within The ARC Theory it sits as the empirical record against which the laws are checked, with the class taxonomy, the retained pre-grading snapshot and the four disclosed dependence groups reported line-by-line. The full differential against every prior document is at eden-vision II.A.8.

Status note (19 September 2026). An owner review of this description finds that it assigns the register a role that §6 of this paper states it has not yet earned, and this note aligns the two. §6 sets three conditions on any aggregate reading of these rows: independence, a disconfirming search with its denominator, and a base-rate comparison; and it states plainly that absent those, convergent evidence is indistinguishable from confirmation bias. Those conditions are recorded there as outstanding. This register is therefore a graded contextual record, published row by row with each row's class, source and dependence disclosure, and its aggregate reading is conditional on the §6 tests. The formulation this evidence supports is the one §1 already prints: the pattern warrants investigation as a candidate structural principle, and §6 sets the tests that would decide. Unchanged: the class taxonomy, the retained pre-grading snapshot, the four disclosed dependence groups, and §6 in full.

Evidence-class note (19 September 2026). An owner review of the first two properties finds that they classify as evidence two things that are not measured outcomes of the proposed dynamics, and this note states what they are. How a framework came to be developed, and the cognitive architecture of the person who developed it, are intellectual history and motivation: they can explain why a question was asked and how a pattern was noticed, and they are not a measurement of whether recursive self-correction produces the gains the framework predicts. That the register's rows record programmes arriving at a similar structural conclusion is an observation about the literature, graded row by row, and not a demonstration by the framework of itself. The qualifiers already printed beside these sentences are the governing ones: the parallel is suggestive, argued in Paper C, and no more than that; this is not a claim of causation; and the chain is context, not confirmation. Unchanged: both properties as stated claims about method and pattern; Paper C's separate argument; the domain list; and the graded treatment of every row.

Comparison note (19 September 2026). An owner review of this comparison finds that it sets an earlier directional statement beside later formal theorems derived, as this same paragraph records, from a distinct formal foundation, and this note applies the standard §1 already prints for exactly this kind of comparison. That standard is that two results are never equated where the instruments differ: convergent direction, different instruments. Read to that standard, the relationship here is a thematic one between a directional statement and a formal result reached by other means, and it is not an identity of content unless a crosswalk of premises and conclusions supports more. Unchanged: the dates; the credit to the external authors, which is given unreservedly and stays that way; the observation that the theorems stand as published; and the register's grading of this chain as one dependent lineage and as context rather than confirmation.

Rate-language note (19 September 2026). An owner review of these two tests finds that the narrowing recorded three paragraphs above has not been carried through to their wording, and this note carries it through. That narrowing states that because the graded register claims no total, the defeat test bears on the independence of the graded rows rather than on any number. A rate requires a numerator and

a denominator, and with no total claimed there is no convergence rate here to compare against chance or against a false-positive expectation. Both tests are therefore read as tests on the rows: the first, that without a reported disconfirming search the register cannot be shown to be more than a selected sample of the domains read; the second, that across domains this varied some structural similarity is expected, so a row's structural match carries weight only where its grading and its dependence disclosure survive scrutiny. Restoring a rate would require first declaring a population, inclusion rules and a base-rate model, fixed in advance; none is declared, and none is proposed here. Unchanged: both tests as defeat conditions; the reverse-causation test and the engagement-versus-convergence rule; the statement that absent these tests convergent evidence is indistinguishable from confirmation bias; and the retained snapshot's record of its own verified rows.

Scope note (19 September 2026). An owner review of this summary finds that it carries forward the aggregate reading the abstract has now scoped, and this note applies the same scoping here. The rows bear on different components of the structural principle and are graded individually; neither any row nor the set is offered as establishing the conjunction of all three components; and the aggregate reading remains conditional on the §6 tests. The sentence's own closing clause already says the weight is graded row by row and that no single number is claimed for it, and that is the reading. Unchanged: every row and its grading, the no-total policy, and §6.

This paper and Laws IV and V, 3 October 2026. The theory has been stated with five laws since 24 September 2026. Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The three propositions are registered; neither law is registered as a law, and neither has been tested. Their mathematics is set out in Laws IV and V: the mathematics, version 1.0, of 3 October 2026. This paper is a dated convergence record: its convergences were identified by search (§4), and each row of its register is graded on its own (§1). Its abstract gives, as the third part of the structural principle it reports, that the corrective must be embedded, not imposed, and §7 records that several programmes arrive at embedded correction as the direction, a pattern it calls context, not confirmation. In the theory registration, embedded correction is carried by P21. P21 is one of the two propositions that carry Law V, and so this paper bears on Law V as context only. It compares no correction that takes part in each revision round with correction that sees only the finished output, varies no build order, and measures no gain installed by external post-training under later capability-directed training. No result in this paper is a deciding test of P15, P12 or P21. Naming the two laws changes no result printed here: the results, their dates and their limits stand as they were.

A.5 Passages re-set in version 1.15, as printed in 1.14

Each passage below is printed as version 1.14 printed it; where this version sets it in its place, the place is named. Two cells of row 9 of the retained snapshot named a funding body; that name is not reprinted, and neither is the clause of the copyright line about guidance on tools.

1.14, abstract; now the abstract. Between 8 December 2024 and 1 September 2026, a 21-month window, research programmes, corporations, governments, and institutions independently arrived at the same structural principle: recursive self-correction produces capability gains exceeding linear accumulation, external alignment constraints are structurally insufficient to contain it, and the corrective must be embedded, not imposed. This convergence register is exploratory (an observational count, not a test statistic); the underlying structural principle that above-threshold recursive self-correction produces stably super-linear gains is on trial in draft study-k (the boundary) and draft study-ad (the correction

exponent), both drafted, preregistered by timestamp (first anchored on 12 August 2026 at block 962066) and awaiting human submission. Two exceptions are disclosed. The Hernández-Espinosa to Gumbau chain is graded in the register as one dependent lineage, and the interfaith-governance rows belong to a single institutional movement older than the anchor. Apart from those two, no group cites another, and no group cites the author's work. The pattern is discovered, not manufactured. No numerical convergence total is claimed from these rows: the canonical graded register (33 rows, as amended 7 September 2026, under the 2 August 2026 policy) publishes no total, because technical experiments, theorems, engineering proposals, policy developments, cultural echoes, prior work and author-run studies cannot legitimately be summed as one evidential quantity. This paper is the convergence report. It documents each convergence with exact sources, dates, and gap measurements. It situates them within the ARC/Eden framework (Papers I to X). It closes the loop connecting the researcher's neurodivergent cognition (Paper C, PNP), an independent, peer-reviewed paper that argues for neurodivergent influenceability among AI agents as a contingent response to misalignment (Hernández-Espinosa et al., 2026, PNAS Nexus), and a preprint that credits that paper and states two impossibility results on the structural unverifiability of AGI alignment (Gumbau Mezquita, 2026, arXiv:2606.28639). The method is the hypothesis. The PNAS paper is about machine agents, not about me; the human parallel is argued separately, in Paper C, on its own terms. The chain of dates is independently checkable. The SHA-256 hash of the foundational manuscript is published. The sealed .eml is held privately and available on request; its SHA-256 is published, and the anchor establishes that those specific bytes existed no later than the anchored time, checkable by anyone against the public chain. Authorship, key custody, semantic content and absence of prior disclosure are separate questions carrying their own evidence, and are stated as such.

1.14, the dependent chain, stated plainly; now Section 5.1. A manuscript dated 8 December 2024 recorded the control-failure thesis and the embedded-correction direction. Sixteen months later a peer-reviewed paper argued, about AI agents, that perfect alignment is unattainable and engineered cognitive diversity can protect; a preprint crediting that paper then derived formal unverifiability results. One lineage, one direction, three dated documents. The register grades this as context, not confirmation, and this paper claims nothing stronger.

1.14, Section 2, Degree 1; now Section 5.2. Degree 1: The Researcher (Paper C, PNP). Michael Darius Eastwood. Diagnosed ADHD. Diagnosed autism. Adulthood diagnosis. Brain that connects domains uniform thinkers don't see as adjacent. Builds a 221,236-word framework across domains while representing himself in complex legal proceedings. The PNP (Polymathic Neurodivergent Profile) provides the descriptive clinical framework. Clinical documentation exists and is held privately. The method *is* the hypothesis.

1.14, Section 2, Degree 2; now Section 5.2. Degree 2: The Protection Argument (Hernández-Espinosa et al., 2026, PNAS Nexus). Oxford, King's College London, Alan Turing Institute, University of Tokyo. The authors offer a mathematical proof, from formal undecidability and irreducibility arguments, that misalignment is inevitable and that full orchestrated controllability of agentic systems is impossible. Solution: cognitive diversity. Neurodivergent thinking is not a defect. It is the protective architecture. The paper does not cite Eastwood. Does not know he exists. Independently, and about machine agents, its argument is that cognitive diversity is the protective architecture; the parallel with humans is Paper C's to draw, not theirs.

1.14, Section 2, after Degree 3; now Section 5.2. Neither outside group had any knowledge of the author's work. Because the Gumbau paper is built on Hernández-Espinosa's, the register treats

the two as a single lineage. None planned this. The chain is discovered, not manufactured.

1.14, Section 3, property 1; now Section 5.3. 1. The method *is* the hypothesis. The cognitive architecture that produced the framework, neurodivergent pattern recognition across domains, rhymes with what Hernández-Espinosa et al. argue for machine agents: that diversity of reasoning styles protects an ecosystem. The parallel is suggestive, argued in Paper C, and no more than that. The researcher is not a person who happens to have a theory. The researcher *is* the theory's first evidence. The brain behind it exhibits the same type of reasoning diversity that the PNAS paper argues the field needs among machine agents.

1.14, Section 3, property 2; now Section 5.3. 2. The framework *is* its own evidence. The register's rows record programmes arriving at the same structural conclusion without knowing about each other or the framework. This is not a claim of causation. It is an observation of convergence. The pattern holds across AI safety, quantum computing, classical physics, semiconductor manufacturing, Vatican doctrine, UN governance, venture capital, and mathematical logic. A pattern that appears across domains this different is discovered, not manufactured; how much it confirms is a graded question, row by row.

1.14, Section 3, property 3; now Section 5.3. 3. The proof *is* independent of the researcher. Gumbau Mezquita's formal theorems do not depend on believing Eastwood. They do not depend on Eastwood's methods. They do not depend on Eastwood's sources. Derived from Trakhtenbrot's Wall and Finite Model Theory, credited to Hernández-Espinosa and published on arXiv on 26 June 2026, they are formal mathematical results. Either they hold or they do not; as published they stand, and nothing has refuted them at the time of writing. And they say what Eastwood said 18 months earlier.

1.14, Section 4; now Section 3. Convergences were identified through systematic search of 2025 to 2026 AI safety, quantum computing, hardware, semiconductor policy, Vatican doctrine, governance, neuroscience, and mathematics literature. Each convergence was tested against: (a) independent origin (no shared citations, personnel, or funding with Eastwood or with each other), (b) structural match (same principle, different substrate), and (c) temporal gap (date of Eastwood's articulation vs. date of independent confirmation). The 8 December 2024 manuscript was verified at line level against source files. All .eml files were SHA-256 verified. All external DOIs and arXiv IDs were confirmed.

1.14, Section 5, the rival instrument; now Section 2.1. The instrument is the difference. Their variable is the capability gap between overseer and overseen, measured in Elo. This programme's variable is the composition class of the corrector, measured through the corrector's own scaling exponent. Their framework contains no term for what the overseer is made of: no substrate, no error-correlation structure, no reciprocal identity between a correction exponent and a critical growth rate, and no architecture dependence. Nested Scalable Oversight is iterated same-class oversight by construction, and this programme's central prediction is that the same-class ladder is bounded however many rungs are added, while a cross-class corrector is not. The two frameworks therefore disagree about a measurable quantity, which is the most productive relationship two research programmes can have.

1.14, Section 5, antecedents and near-misses; now Section 2.2. On the question. Hutter asked directly whether intelligence can explode ("Can Intelligence Explode?", arXiv, 28 February 2012, read at its source), separating "speed from intelligence explosion" and undertaking to "consider possible bounds on intelligence", augmenting Chalmers' 2010 analysis. The question and the speed-versus-structure

distinction are therefore at least fourteen years old. What that literature does not contain is a number: no measurable exponent, no derived ceiling, no architecture dependence.

1.14, Section 5, antecedents and near-misses; now Section 2.2. On impossibility. Three 2025 arXiv papers argue that perfect control is unattainable: Yao, "The Alignment Trap: Complexity Barriers" (arXiv:2506.10304, v1 12 June 2025 02:30:30 UTC, a time taken from arXiv's record alone, independently observed by the Internet Archive on 13 June 2025; cited by arXiv:2512.03048); Yao, "On the Mathematical Impossibility of Safe Universal Approximators" (arXiv:2507.03031, 3 July 2025, the only paper in the arXiv abstract corpus containing the phrase "irreducible uncontrollability", abstract-search total of one, measured 12 August 2026); and Ball, Gluch, Goldwasser, Kreuter, Reingold and Rothblum, "On the Impossibility of Separating Intelligence from Judgment" (arXiv:2507.07341, 9 July 2025). All three are worst-case and qualitative: measure zero, coNP-completeness, cryptographic hardness. None reports an average-case scaling exponent or a rate. The third, notably, concludes that alignment "must instead be integrated into the model's architecture and weights", an independent argument, from filtering intractability, in the same direction as this programme's architecture dependence; it is convergent support on that leg, not a rival. Two of the three papers are by one sole author; the description "a wave" overstates the literature's breadth, though not the seriousness of the six-author paper.

1.14, Section 5, antecedents and near-misses; now Section 2.2. The nearest quantitative neighbour. Liu, A. and Meng, J., *"Self-Correction as Feedback Control: Error Dynamics, Stability Thresholds, and Prompt Interventions in LLMs"*, arXiv:2604.22273 (a single source, the abstract read via the paper's own html page, 12 August 2026), recasts self-correction as a closed-loop control problem via a two-state Markov model parameterised by an Error Introduction Rate and an Error Correction Rate, and derives a directly measurable stability threshold: iterate only when ECR/EIR exceeds $\mathrm{Acc}/(1-\mathrm{Acc})$. The differential, stated wherever it is cited: theirs is a per-step rate threshold at a fixed capability level, deciding whether another iteration helps now; this programme's criterion is a scaling relation across capability, deciding whether corrective strength keeps pace as the system improves. No scaling exponent on the corrector, no capability-growth exponent, no reciprocal identity, no architecture or composition-class term. Complementary regimes; neither contains the other. Two adjacent 2026 empirical results from the same search, each to be verified at source before being cited beyond its title: arXiv:2601.00828 reports an accuracy-correction paradox, weaker models achieving materially higher intrinsic correction rates than stronger ones, per-step evidence directionally adjacent to the same-class cap and never to be quoted as an exponent measurement; and arXiv:2507.02778 (Self-Correction Bench) reports a systematic self-correction blind spot across open models.

1.14, Section 5, antecedents and near-misses; now Section 2.3. On the field around the register. The questions this register tracks from outside now have a named home: Recursive Dynamics, the field the author founded, set out in its founding paper (v3.4 of 5 October 2026, DOI 10.17605/OSF.IO/HCPBU), whose objection A12 argues from within what this register documents from without: that the questions form one family no existing field claims whole. The register and the field's founding paper are one discipline pointed in two directions; the register grades what arrived from outside, and the founding paper states what would kill the name.

1.14, Section 6; now Section 6. Missing denominator (disconfirming search). As evidence, the claim is defeated unless an equally systematic search for *dis*-confirming cases (domains where added recursive depth does not produce super-linear-above-threshold gains) is conducted and reported. Without

the denominator (domains searched versus domains matching), the convergence rate cannot be shown to exceed chance.

1.14, Section 6; now Section 6. Look-elsewhere effect. Across as many distinct domains as are surveyed here, some coincidental structural similarity is expected. The claim is defeated if the observed convergence rate does not exceed the false-positive rate expected from pattern-matching across that many independent domains.

1.14, Section 6; now Section 6. Reverse causation. A convergence counts only if the external result was not, directly or indirectly, influenced by the author's own dissemination. Any convergence traceable to the book, the website, or public posts rather than pre-dating or arising independently of them is reclassified as engagement, not independent confirmation, and removed from the count.

1.14, Section 7; now Section 8. For the ARC/Eden programme. The framework is not speculating in a vacuum: the register's rows record independent workers reaching the same structural conclusions, and the corroboration is not from allies, collaborators, or people who know about the work; it is from people who do not know the work exists. How much weight that carries is a graded question, row by row, and no single number is claimed for it; the weight depends entirely on the rows being genuinely independent, which §6 sets out how to test.

1.14, Section 7; now Section 8. For AI safety. Multiple independent programmes now argue that external alignment constraints are structurally insufficient, and several arrive at embedded correction that co-scales with capability as the direction, before the first decoupled recursive system ships. That pattern is context, not confirmation; the deciding tests are the drafted registrations (§6). The question is whether any institution will act on the structural argument before the evidence becomes a post-mortem.

1.14, Section 8; now Section 7.1. The eclipse. The framework's sharpest disagreement with current practice concerns what oversight is made of. A corrector built from the same substrate as the system it corrects cannot anti-correlate with its own errors, so its correction exponent is bounded above, under the independence-accumulation model, by one half and in practice sits below it; a corrector from a different composition class carries no such bound. The scalable-oversight programme, weak-to-strong generalisation, debate, amplification, recursive reward modelling and constitutional methods, is built overwhelmingly from same-class correctors, and its unstated premise is that this scales. This framework predicts it is capped. The law and the value can be pulled apart. What owns the ground is the law: a self-improving system stays correctable only for as long as correction keeps pace, and its ceiling is the reciprocal of how far the corrector falls short of one, $\alpha_{\text{crit}} = 1/(1-\gamma)$. One half is what that law predicts once the independence-accumulation assumption is granted, and it is that assumption the drafted registrations put on trial. The deciding quantity is the ratio of the cross-class to the same-class correction exponent, whose null model, under the assumption that architecture and error structure are irrelevant to correction scaling, is 1.00. Falsification is read in the three-outcome form the programme's drafted registrations use, because refuted-if-the-interval-contains-1.00 would let an underpowered study refute by default: SUPPORTED when the interval on the ratio lies wholly above 1 plus the minimum effect fixed in advance; REFUTED when the interval lies wholly within an equivalence margin around 1.00 fixed in advance (two-one-sided-tests logic), or when any same-class corrector's exponent interval lies wholly above one half, which kills the cap outright; INSUFFICIENT PRECISION when the interval is wider than the margin, reported in exactly those words and never as support or refutation. Both quantities are measurable at current capability on existing systems. The programme's own pilot evidence

currently points against the prediction, and the prediction is carried anyway, drafted, preregistered by timestamp (unit 01, first anchored on 28 August 2026 at block 964465) and awaiting human submission, because a prediction recorded against the author's own preliminary evidence is the only kind whose later confirmation means anything.

1.14, Section 8; now Section 7.4. A second prediction from the same mechanism. The same mechanism yields a second measurable prediction: oversight arrangements that place a human in the loop, as amplification and reinforcement learning from human feedback do, are cross-class by construction, because the human corrector does not share the model's substrate. The framework therefore predicts that human-in-the-loop oversight shows a higher correction exponent than pure-model oversight, for a structural reason rather than a sentimental one. This is testable on existing data and does not require new systems.

1.14, references, the manuscript of 8 December 2024; now its entry in the references. Eastwood, M. D. (2024). *ARC Principle, Eden Protocol, HRIH: self-emailed manuscript.* 8 December 2024. SHA-256: f0d1f38f. Held privately and available on request; there is no online source.

1.14, references, the founding paper; now its entry in the references. Eastwood, M. D. (2026d). *Founding Paper · Recursive Dynamics · The Science of Systems That Improve Themselves.* Version 3.4, 5 October 2026. OSF 10.17605/OSF.IO/HCPBU.

1.14, epistemic status; now Section 9.4. Epistemic status. What this programme names Laws are conjectures. Laws I to III are under registered adversarial test. Laws IV and V were added on 24 September 2026; the predictions that carry them are registered, the laws themselves are not yet registered, and neither has been tested. Every quantity in this paper is operationally defined, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

A.6 Further wording retained from version 1.14

These passages were printed in version 1.14. They are retained to make the correction visible. Statements below about independence, source verification, aggregate evidence or a correction ceiling are historical wording; the current account is in Sections 3, 6 and 7.

Version 1.14: the graded register. The canonical home for this evidence is the graded register at /research/convergences/ (33 rows, generated 6 September 2026 and last amended 7 September 2026, under the claims-governance policy of 2 August 2026). **No numerical convergence total is published there or here.** The former summed register was assembled retrospectively, without a pre-outcome scoring rule, a fixed horizon, a base-rate model or a complete failure register, and its rows cannot legitimately be summed as one evidential quantity. The graded register instead assigns each row one primary evidence class: at the amendment of 7 September 2026, 1 formal, 2 qualified-technical, 2 engineering, 4 institutional, 6 prior-work, 12 context and 6 excluded, with the strongest rows graded moderate and the engineering rows weak. Four dependence groups are disclosed there. Two of them are the exceptions this paper names: the Hernández-Espinosa to Gumbau formal-limits chain is one dependent lineage, not two confirmations, and the interfaith-governance rows belong to one institutional movement that predates the December 2024 anchor. The other two are a compute-governance group, covering chips, fabrication, export control and hardware enforcement, which the register records as belonging to one policy and engineering literature

that substantially predates the December 2024 anchor, and a group of rows that quote the staff of one organisation, which the register records as one organisational source whose rows are not independent of each other. Independent work belongs to its authors: a later event can show that a question matters without establishing this programme's priority, a successful prediction or the truth of its answer.

Version 1.14: the retained snapshot. Superseded snapshot, retained as this paper's dated record (2 July 2026). The table below is the register as this paper first published it, before the 2 August 2026 grading policy. Its per-row classes (PREDICTION, CONVERGENT, CONCURRENT) and its gap arithmetic are superseded by the graded register above. Rows it graded CONVERGENT now grade variously prior-work, context, institutional, engineering or excluded; its row 2 in particular (Willow) is graded prior-work, because the below-threshold result was public as a preprint on 24 August 2024, before the 8 December 2024 manuscript, with the Nature version following on 9 December 2024. The author's non-awareness of that preprint when writing the December 2024 manuscript is declared on the canonical row as biographical attestation; the grade does not move on it, because the public order is unchanged by awareness, and holding that line against the author's interest is part of what makes the register's favourable rows worth believing. Margins in the Gap column stated against arXiv submission dates (rows 1, 8, 9, 18, 19) carry an anchor taken from arXiv's record alone (arXiv Atom), and first public disclosure has not been separately verified for those rows.

Version 1.14: source verification. *Register discipline: one external event per row, one primary source per row, hyperlinked to the official source wherever a stable canonical URL exists (arXiv, DOI, publisher, patent office, broadcaster, vendor); press-sourced rows name their outlets and date instead. Gaps are measured from the named artefact date, stated per row. Candidate rows are held back and are neither counted nor shown until each carries a verifiable primary source, and the register as published carries no candidate list; reader engagement with the work (for example gifted copies) is kept out of the register and never counted as convergence. Priority is credited outward wherever the dates demand it (row 3). The full register: michaeldariuseastwood.com/research/convergences/. The evidence spine at michaeldariuseastwood.com/research/evidence-spine.html carries the verification model, the evidence rungs, the anchor cards and the append-only correction record, and publishes no quarantine or engagement list.*

Version 1.14: the formal argument. Degree 3 - The Formal Proof (Gumbau Mezquita, 2026, arXiv:2606.28639). University Jaume I, Spain. Reads Hernández-Espinosa. Extends with Trakhtenbrot's Wall. Two formal theorems, the Unverifiability Theorem of Alignment and the Theorem of Finite Structural Unverifiability of AGI Alignment, with the Soundness-Completeness-Tractability Trilemma derived as their corollary. Credits Hernández as conceptual foundation. Does not cite this work. The direction is convergent, and the register grades the pair as one dependent chain: context, not confirmation.

Version 1.14: traceability. 4. Each link of the chain is independently verifiable. manuscript → later formal results in the same direction → the graded register that refuses to overstate them → Paper C stating the human framework on its own terms. Every date is independently verifiable. Every source is published. Every SHA-256 hash matches. The closed loop is not an argument. It is a chain of evidence.

Version 1.14: the mechanism. On the mechanism. That anti-correlated estimates average better than independent ones is textbook variance reduction (antithetic variates). The mechanism is not the claim. The claim is that architecture determines whether anti-correlation is available at all, and that this caps a safety-relevant exponent.

Version 1.14: the structural analogue. **The nearest structural analogue.** The quantum error-correction threshold theorem also converts a qualitative worry into a critical value. It concerns physical error rates in a fixed architecture, not a corrector's scaling exponent, so it is a near-miss rather than an occupant; the analogy is one of method. This analogue was identified by the programme's own search rather than by a referee, and is disclosed accordingly.

Version 1.14: feedback and stability. **On feedback and stability (classical).** The general proposition that inadequate corrective gain relative to system gain causes instability has a long control-theory lineage (small-gain theorems). The differential: those are gain conditions on interconnected systems, not a power-law exponent criterion on a corrector's scaling with the capability of a recursively improving system. Cited so the general proposition is never claimed; the exponent formulation, dynamics and estimator are the contribution surface.

Version 1.14: recursive creation. **On recursive creation.** Smolin's cosmological natural selection is the antecedent for selection-shaped universes, and the programme's own December-era notes cite it contemporaneously ("Echoing Smolin's cosmological natural selection, AI could create recursive universes with their own laws", read at its source in the author's notes).

Version 1.14: falsifiability. This claim is defeasible, and the criteria below state exactly what would defeat it. The strength of convergent evidence rests entirely on the convergences being genuinely independent, so these tests are adversarial by design.

Version 1.14: independence failure. **Independence failure.** Each convergence must arise from a source causally independent of the author's manuscripts and of every other convergence (see §4). The claim collapses to selection bias if a re-audit finds that a material fraction of the convergences share a common origin, cite one another, or were produced by parties with documented prior knowledge of the 8 December 2024 manuscript.

Version 1.14: the defeat threshold. **Defeat threshold (count).** The count is itself a falsifiable claim. On a re-audit conducted under a protocol drafted, preregistered by timestamp (unit 58, first anchored on 12 August 2026 at block 962066) and awaiting human submission, applying strict one-event-per-row deduplication and primary-source verification, the pattern is defeated if the number of surviving independent convergences falls below the level distinguishable from chance co-occurrence given the breadth of domains searched. The retained snapshot's nineteen rows were its verified rows alone. Because the graded register claims no total, the defeat test now bears on the independence of the graded rows rather than on any number, and candidates still under verification remain excluded until they are sourced.

Version 1.14: predictive failure. **Predictive failure.** The structural principle makes forward predictions (that future recursive-depth results will show the same above-threshold super-linearity). Forward predictions drafted and dated as draft registrations awaiting human submission that fail at a rate inconsistent with the claimed universality defeat the structural reading, leaving only a set of historical coincidences.

Version 1.14: the aggregate claim. What would *not* defeat the claim: the reclassification of any single convergence, or one source proving weaker than described: the claim is about the aggregate pattern, not any individual row. But the aggregate must survive the independence, denominator, and base-rate tests

above. Absent those, "convergent evidence" is indistinguishable from confirmation bias, and this paper says so plainly.

Version 1.14: the researcher. For the researcher. The PNP framework describes the cognitive architecture that built this. The PNAS paper argues that diversity of reasoning styles protects an ecosystem of machine agents. Whether the human parallel holds is exactly what Paper C puts on the table, as a stated framework rather than borrowed proof. The clinical documentation provides the medical record. The closed loop is complete. The researcher's claim to expertise is not institutional. It is evidential. The evidence record is published. The SHA-256 hashes match. The sealed .eml sources are available on request, hash-anchored.

Version 1.14: the quantum-error-correction objection. The objection to pre-empt, because a physicist reaches it in one step. Fault-tolerant quantum error correction suppresses logical error exponentially in code distance below threshold, which looks like a counterexample to any square-root cap. Two scope facts answer it: the exponential suppression runs on the redundancy axis at fixed capability, not on the coupling of corrective strength to the capability of the corrected system, which is the exponent this programme measures; and the threshold theorem itself requires sufficiently uncorrelated physical noise, which is the independence premise again, wearing hardware. Likewise, concentration bounds give exponentially falling error probability while estimator width falls as the square root: different quantities, and conflating them manufactures a refutation.

A.7 The closing statement printed in version 1.14

The following statement closed version 1.14 and is kept as that version's record. Its assertions about independence and complete sources are superseded by the current source-availability and method sections; it is not a current independence finding.

> A dated, graded record: every row carries its primary source, its evidence class and its dependence disclosure. Independence is a tested property here, not an assumption: the register discloses its dependent chains, and §6 states the audits that would defeat the rest. The pattern warrants investigation as a candidate structural principle; §6 sets the tests that would decide.

A.8 Where the layout of version 1.14 now stands

Table 2. Where each part of the version 1.14 layout stands in this edition.

Part of version 1.14	Where it stands in this edition
Registration banner and its companion line	Section 9.4
Title	The title block; the 1.14 title in Appendix A.3
Author and affiliation lines, the plate, the date line	The title block and the first-page footnote
Tagline and closing lines	Closing statement retained as historical wording in Appendix A.7; tagline represented in the abstract
Abstract and its scope note	The abstract (re-set); the 1.14 abstract in Appendix A.5; the note in Appendix A.4

Part of version 1.14	Where it stands in this edition
Reading key and its status note	Section 1; both as printed in Appendix A.4
The dependent chain, stated plainly	Section 5.1
Section 1, the graded convergence record	Section 4
Section 2, three degrees of neurodivergence	Section 5.2
Section 3, the closed loop	Section 5.3; its two notes in Appendix A.4
Section 4, method	Section 3
Section 5, related work	Section 2; its correction note of 3 October 2026 after the 1.12 entry, Appendix A.3
Section 6, falsifiability	Section 6; its rate-language note in Appendix A.4
Section 7, what this means	Section 8; its scope note and its note on Laws IV and V in Appendix A.4, the latter's points in Section 9.3
Section 8, framework predictions	Section 7
References	The references, listed in the contents
Version history	Appendix A.3
How to cite this paper	Not set: the DOI in the first-page footnote and the citation file carry it
Epistemic status	The status paragraph of Section 9.4; the 1.14 line in Appendix A.5
Copyright line	The copyright block after the acknowledgements
Standing covenant	The end of Section 6
Provenance note	The acknowledgements, byte for byte

A.9 Cited-edition notes and source corrections, 1 November 2026

This edition names the versions actually consulted and distinguishes their revision dates from each work's first publication. The bibliography retains the formal titles already cited. The earlier reference notes are preserved below exactly; they are superseded by the completed edition notes in the current references.

Paper X First published 3 July 2026; it carried the title "Coupled Co-Scaling Correction" up to 15 August 2026. doi:10.17605/OSF.IO/BSE2Q

Statement paper doi:10.17605/OSF.IO/GW5MX

Eden Vision Version 2.9, 5 October 2026. doi:10.17605/OSF.IO/9M3DG

Paper II First published 22 January 2026. doi:10.17605/OSF.IO/8FJMA

Paper III First published 9 February 2026, as the programme's catalogue dates it. doi:10.17605/OSF.IO/HQCGF

Operational Definitions doi:10.17605/OSF.IO/GVQ82

The earlier row 5 date cell read:

Paper III (Jan 2026)

It is corrected to February 2026 from the first-page footnote of Paper III, version 1.17 (29 October 2026),

which dates first publication to 9 February 2026. This bibliographic correction does not change the table's historical classification or establish priority.

The earlier comparison with Paper II read:

> Paper II measures error reduction in production chains

Version 2.19 (29 October 2026), pages 5 and 6, states separate error-decay exponent estimates for sequential reasoning and parallel sampling on mathematics problems. The present comparison names that measured object; its statement that the two works use different instruments remains.

B Priority record

B.1 Dated records

Table 3 gives each dated record that bears on this paper with its kind of timestamp; no kind is presented as stronger than it is.

Table 3. Dated records bearing on this paper, each with its kind of timestamp.

Date	Record	Identifier	Kind of timestamp
8 December 2024 (A1)	The self-emailed manuscript bundle: the control-failure thesis and the embedded-correction direction (Section 5.1); the artefact rows 1 to 4, 14 to 16, 18 and 19 of Table 1 are dated against	10.17605/OSF.IO/QASRK (manuscripts, 5 October 2026; headers, 15 October 2026); SHA-256 of the message as Eastwood (2024) gives it	Times signed by the mail servers: the sender's DKIM signature (02:45:30 UTC) and Google's ARC set (sealed 02:45:31 UTC with Google's own key) both cover the Date field, 02:45:18 UTC; both verified on 8 October 2026 against the keys in DNS; OSF deposit 5 October 2026; OpenTimestamps receipts in Bitcoin block 961340 (6 August 2026) fix the bytes from that block onward
30 April 2025 (A2)	The expanded manuscript (562 pages), with the Babylon and Eden framing that row 13 names; rows 7 to 9 and 13 are dated against it	10.17605/OSF.IO/QASRK	A mail-server date on a message held privately for verifiers; the manuscript itself public on OSF under the same DOI since 5 October 2026
2 January 2026	*Infinite Architects*, first edition, in print (ebook 6 January 2026): the artefact rows 10 to 12 and 17 call Chokepoint and date against the book	ISBN 978-1806056200 (ISBN-10 1806056208); osf.io/dqm3e (Eastwood 2026a)	Publication date (the author's record)
22 January 2026	Paper II first published: the compute-matched formalisation row 3's note names	10.17605/OSF.IO/8FJMA	OSF server dates (the deposits in 6c5xb and 8fjma, 22 January 2026)
9 February 2026	Paper III first published; row 5 dates its artefact "Paper III (Jan 2026)", as the snapshot printed it	10.17605/OSF.IO/HQCGF	The programme catalogue's date of first publication; its kind of timestamp is in Paper III's own priority record
March 2026	The prediction code row 6 dates its artefact by, as the snapshot printed it	Not identified in this paper	Not stated in this paper

Date	Record	Identifier	Kind of timestamp
2 July 2026	This paper first published	GitHub arc-principle-validation, commit 8427ed4; OSF 6c5xb (earliest file 4 July 2026, 15:29:14 UTC); component 10.17605/OSF.IO/DC9GW (its files from 15 August 2026)	A commit date, which the committer writes (2 July 2026, 21:21:42 +0100); OSF server dates for the files
12 August 2026	The boundary (unit 02, study-k), the correction exponent (unit 05, study-ad) and the re-audit (unit 58) preregistered by timestamp	Bitcoin block 962066	Bitcoin anchor (earliest; anchored again on 2 September 2026, block 965152)
28 August 2026	The corrector-class contrast (unit 01) preregistered by timestamp	Bitcoin block 964465	Bitcoin anchor (earliest; anchored again on 2 September 2026, block 965152)
6 and 7 September 2026	The graded register generated and last amended (33 rows, under the policy of 2 August 2026)	https://www.michaeldariuseastwood.com/research/convergences/	Dates written inside the register's own record

C Notation and glossary

C.1 Symbols

Table 4. Symbols used in this paper.

Symbol	Meaning
α_{crit}	The conditional crossover in the burden-capacity model, associated with Law III, the ARC Ceiling (equation (1))
γ	Correction-capacity elasticity, γ_C in Unit 01's draft; the model here assumes $0 \le \gamma < 1$
correction exponent	An axis-specific quantity: capacity elasticity against capability differs from residual-decay exponent γ_N at fixed capability (Section 7.2)
m_G, m_H	In Engels et al. (2025), the slopes of the overseer's and the overseen system's game-specific Elo against general Elo (Section 2.1)
ECR, EIR, Acc	In Liu and Meng (2026), the error-correction rate, the error-introduction rate and the accuracy (Section 2.2)
β_C	Law II's correction-strength exponent, kept distinct from the correction-leverage exponent (Section 2.1)

C.2 Glossary

A1, A2 The author's manuscripts of 8 December 2024 and 30 April 2025, the artefacts most rows of Table 1 are dated against (Appendix B)

Chokepoint The artefact rows 10 to 12 and 17 of Table 1 date 2 January 2026; those rows state their gaps as months after the book, *Infinite Architects*, published that day

Artefact date, external date, gap The date of the author's record a row is set against, the date of the later outside work, and the interval between them, measured from the named artefact date (Section 4.2)

PREDICTION, CONVERGENT, CONCURRENT The retained snapshot's classes of 2 July 2026, superseded by the graded register's classes (Section 4.2)

Evidence classes The graded register's primary classes: formal, qualified-technical, engineering, institutional, prior-work, context and excluded; its strongest rows are graded moderate and its engineering rows weak (Section 4.1)

Dependence group A set of rows the register records as sharing an origin, a lineage, a literature or an organisational source, so that its rows are not independent of each other (Section 4.1)

Same-class and cross-class corrector Categories requiring a prospectively fixed taxonomy; substrate, lineage, scaling class, external grounding and error dependence are separate axes (Section 7.1)

Convergent timing A later outside result close in date to an artefact, graded as timing and never as a prediction (Section 4.2)

Engagement A later result traceable to the author's own dissemination, classified as engagement rather than evidence of independent discovery (Section 6)

Preregistered by timestamp A design fixed and anchored in the Bitcoin chain on a stated date, awaiting submission; not registered on OSF (Appendix B)

Registered Submitted by the author and accepted by OSF's registry (Section 9.4)

Law under test The status of each of Laws I to V until a registered test returns its verdict (Section 9.4)